

Conical Pendulum 2

Linking Concepts in Pre-Uni Physics 18.2

Subject & topics: Physics | Mechanics | Circular Motion

Stage & difficulty: A Level P2

A small sphere of mass 2.0 kg, attached to the end of a light string of length 90 cm at 24° to the vertical, moves in a horizontal circle.

Part A

Tension

Calculate the tension T in the string.

Part B

Height above position at rest

Calculate the height h by which the mass is raised above its position at rest.

Question deck:

STEM SMART Physics 24 - Applied Circular Motion

Conical Pendulum 3

Linking Concepts in Pre-Uni Physics 18.3

Subject & topics: Physics | Mechanics | Circular Motion **Stage & difficulty:** A Level P2

A lead ball of mass 45 g is attached to the end of an 80 cm long light string and swung around in a horizontal circle at high speed.

If the string snaps at a tension of 195 N, what is the maximum frequency of rotation f possible?

Question deck:

STEM SMART Physics 24 - Applied Circular Motion

Elastic Circular Motion

Subject & topics: Physics | Mechanics | Circular Motion **Stage & difficulty:** A Level C2

A mass, m of 0.050 kg is attached to one end of a piece of elastic of unstretched length, $l = 0.50\text{ m}$. The force constant, k of the elastic (i.e. the force required to produce unit extension) is 40 N m^{-1} . The mass is rotated steadily on a smooth table around a fixed point in a horizontal circle of radius $r = 0.70\text{ m}$.

What is the approximate speed of the mass?

- ☐ 20 m s^{-1}
- ☐ 11 m s^{-1}
- ☐ 5.7 m s^{-1}
- ☐ 15 m s^{-1}
- ☐ 2.4 m s^{-1}

Adapted with permission from UCLES, A Level Physics, June 1987, Paper 1, Question 9.

Question deck:

STEM SMART Physics 24 - Applied Circular Motion

Conical Pendulum 7

Linking Concepts in Pre-Uni Physics 18.7

Subject & topics: Physics | Mechanics | Circular Motion **Stage & difficulty:** A Level C2

An aircraft travelling at 160 knots maintains its altitude during a circular banked "rate one turn", which is a 3.0° s^{-1} turning rate. (1 knot = 0.514 m s^{-1})

At what angle to the horizontal are the wings of the plane?

Question deck:

STEM SMART Physics 24 - Applied Circular Motion

Vertical Circles 1

Linking Concepts in Pre-Uni Physics 19.1

Subject & topics: Physics | Mechanics | Circular Motion

Stage & difficulty: A Level P3

Quantities:

u speed at bottom (m s^{-1})

v speed at top (m s^{-1})

W weight (N)

r radius of circle (m)

m mass (kg)

N normal reaction (N) [+ means ↑]

a centripetal acceleration (m s^{-2})

F resultant force (N)

Equations:

$$F = ma \qquad W = mg \qquad a_{\text{top}} = \frac{v^2}{r} \qquad a_{\text{bottom}} = \frac{u^2}{r}$$

Gain in $E_{\text{GP}} = \text{Loss in } E_{\text{K}}$, so

$$mg \times 2r = \frac{1}{2}mu^2 - \frac{1}{2}mv^2$$

An object travels in a vertical circle. Using the equations above, and writing upwards normal reactions as positive, write an equation for:

Part A

N (bottom) using W, m, a

N for the mass at the bottom using W, m and a .

The following symbols may be useful: N, W, a, m

Part B

N (bottom) using m, r, g, u

N for the mass at the bottom using m, r, g and u .

The following symbols may be useful: N, g, m, r, u

Part C

N (top) using W, m, a

N for the mass at the top using W, m and a .

The following symbols may be useful: N, W, a, m

Part D

N (top) using m, r, g, v

N for the mass at the top using m, r, g and v .

The following symbols may be useful: N, g, m, r, v

Part E

N (top) using m, r, g, u

N for the mass at the top using m, r, g and u .

The following symbols may be useful: N, g, m, r, u

Part F

v if $N = 0$

the speed v needed at the top if $N = 0$.

The following symbols may be useful: N, g, m, r, v

Part G

u if $N = 0$ at top

the speed u needed at the bottom if $N = 0$ at the top.

The following symbols may be useful: N, g, m, r, u

Question deck:

STEM SMART Physics 24 - Applied Circular Motion

Vertical Circles 2

Linking Concepts in Pre-Uni Physics 19.2

Subject & topics: Physics | Mechanics | Circular Motion

Stage & difficulty: A Level P3

Calculate the normal reaction when a 1200 kg car is half way over a hump back bridge if it is travelling at 8.0 m s^{-1} . The radius of the bridge's arc is 23 m.

Question deck:

STEM SMART Physics 24 - Applied Circular Motion

Vertical Circles 3

Linking Concepts in Pre-Uni Physics 19.3

Subject & topics: Physics | Mechanics | Circular Motion

Stage & difficulty: A Level P3

A 1200 kg car is half way over a hump back bridge. The radius of the bridge's arc is 23 m.

Calculate the speed at which the wheels would just leave the ground at the top of the bridge.

Question deck:

STEM SMART Physics 24 - Applied Circular Motion

Vertical Circles 8

Linking Concepts in Pre-Uni Physics 19.8

Subject & topics: Physics | Mechanics | Circular Motion **Stage & difficulty:** A Level P3

When roller-coaster riders describe their rides, they call the ratio N/mg , that is, normal reaction / weight, the *g-force*. (This is not a scientific term). In this formula, N is taken as positive if it is directed upwards through the rider's body towards their head. A roller-coaster is designed to give $N/mg = 2.5$ at both the top and the bottom of the ride. The loop is not circular. The rider sits in a train which runs around the inside of the loop. The top of the loop is curved with a 7.6 m radius.

Part A

Rider at rest

State the value of N/mg for a rider sitting at rest in the train.

Part B

Speed at top

Calculate the speed of the train at the top of the loop.

Part C

Speed at bottom

If there is no friction, and the top of the loop is 21 m above the bottom, how fast will the train travel at the bottom of the loop?

Part D

Radius at bottom

Calculate the radius of the loop at the bottom of the track.

Question deck:

STEM SMART Physics 24 - Applied Circular Motion